Gabriel Wallin, PhD

Email: g.wallin@lancaster.ac.uk
https://www.lancaster.ac.uk/maths/people/gabriel-wallin
Personal website: http://gabrieltwallin.github.io
LinkedIn: linkedin.com/in/gabrielwallin/

School of Mathematical Sciences
Lancaster University, UK

Academic positions

Lancaster University — Lancaster, UK
Lecturer in Statistics (UK equivalent to tenure-track Assistant Professor),
School of Mathematical Sciences. — 2023–Current

The London School of Economics and Political Science — London, UK
Postdoctoral researcher and Visiting Teacher, Department of Statistics. — 2021–2024

Université Côte d'Azur & Inria — Nice, France
Postdoctoral Researcher, Department of Mathematics. — 2020–2021

Education

Umeå University — Umeå, Sweden
P.hD. in Statistics. — 2015–2020

The London School of Economics and Political Science — London, UK
Visiting Research Student in Statistics. — 2019

Umeå University — Umeå, Sweden
M.Sc. in Statistics. — 2014–2015

Umeå University — Umeå, Sweden
B.Sc. in Statistics. — 2011–2014

Research interests

- **Latent Variable Modeling**
  Latent trait models, exploratory factor analysis, latent class analysis.
- **Statistical machine learning algorithms for quantitative social science research**
  Proximal algorithms, statistical learning with sparsity for factor models.
- **Applications**
  Educational and psychological assessments, public opinion, poverty and inequality studies, humanitarian decision-making.

Publications in peer-reviewed journals

1. **Wallin, G.**, & Wiberg, M. (2026). Smoothing of Bivariate Test Score Distributions: Model Selection Targeting Test Score Equating. *Journal of Educational and Behavioral Statistics*, 51(1), 3–37.

2. **Wallin, G.**, & Wiberg, M. (2026). Propensity Score Methods for Local Test Score Equating: Stratification and Inverse Probability Weighting. *Journal of Educational and Behavioral Statistics*, in press.

3. **Wallin, G.**, Chen, Y., Lee, Y-H., & Li, X. (2025). A latent variable model with change-points and its application to time pressure effects in educational assessment. *The Annals of Applied Statistics*, 19(3), 2490–2516.

4. Laukaitytė, I., **Wallin, G.**, & Wiberg, M. (2025). Combining propensity scores and common items for test score equating. *Applied Psychological Measurement*, in press. (Shared first author).

5. Lunn, J., Ryan, C., **Wallin, G.**, Day, L., & Bailey, A. (2025). Child health inequalities in sleep duration. *European Journal of Public Health*, 35.

6. **Wallin, G.**, Chen, Y., & Moustaki, I. (2024). DIF analysis with unknown groups and anchor items. *Psychometrika*, 89(1), 267–295.

7. Liu, X., **Wallin, G.**, Chen, Y., & Moustaki, I. (2023). Rotation to sparse loadings using L^p losses and related inference problems. *Psychometrika*, 88(2), 527–553.

8. **Wallin, G.**, & Wiberg, M. (2023). Model misspecification and robustness of observed-score test equating using propensity scores. *Journal of Educational and Behavioral Statistics*, 48(5), 603–635.

9. Altintaş, Ö., & **Wallin, G.** (2021). Equality of admission tests using kernel equating under the non-equivalent groups with covariates design. *International Journal of Assessment Tools in Education*, 8(4), 729–743.

10. **Wallin, G.**, Wiberg, M., & Häggström, J. (2021). How important is the choice of bandwidth in kernel equating? *Applied Psychological Measurement*, 45(7–8), 518–535.

11. **Wallin, G.**, & Wiberg, M. (2019). Kernel equating using propensity scores for nonequivalent groups. *Journal of Educational and Behavioral Statistics*, 44(4), 390–414.

Peer-reviewed conference papers

1. González, J. & **Wallin, G**. (2021). An Illustration on the Quantile-Based Calculation of the Standard Error of Equating in Kernel Equating. In Wiberg, M., Molenaar, D., González, J., Böckenholt, U. & Kim, J-S. (Eds.) (2021). *Quantitative Psychology: The 85th Annual Meeting of the Psychometric Society, virtual, 2021*. New York: Springer.

2. **Wallin, G**. & Wiberg, M. (2021). Equating Nonequivalent Groups using Propensity Scores – Model Misspecification and Sensitivity Analysis. In Wiberg, M., Molenaar, D., González, J., Böckenholt, U. & Kim, J-S. (Eds.) (2021). *Quantitative Psychology: The 85th Annual Meeting of the Psychometric Society, virtual, 2021*. New York: Springer.

3. **Wallin, G** (2020). Extensions of the Kernel Method of Test Score Equating. Statistical Studies, 55. *PhD thesis*. URL: https://umu.diva-portal.org/smash/get/diva2:1378833/FULLTEXT01.pdf.

4. **Wallin, G**. & Wiberg, M. (2020). Model Selection for Presmoothing of Bivariate Score Distributions in Kernel Equating. In Wiberg, M., Molenaar, D., González, J., Böckenholt, U. & Kim, J-S. (Eds.) (2020). *Quantitative Psychology: The 84th Annual Meeting of the Psychometric Society, Santiago, Chile, 2019*. New York: Springer.

5. **Wallin, G**., Häggström, J. & Wiberg, M. (2018). How to select the bandwidth in kernel equating - An evaluation of five different methods. In Wiberg, M., Culpepper, S., Janssen, R., González, J., & Molenaar, D. (Eds.) (2018). *Quantitative Psychology: The 82nd Annual Meeting of the Psychometric Society, Zurich, Switzerland, 2017*. New York: Springer.

6. **Wallin, G**,. & Wiberg, M. (2017). Non-equivalent groups with covariates design using propensity scores for kernel equating. In L. A. van der Ark, M. Wiberg, S. A. Culpepper, J. A. Douglas, & W.-C Wang (Eds.) (2017), *Quantitative psychology – 81st annual meeting of the psychometric society, Asheville, North Carolina, 2016*. New York: Springer.

Edited volumes

1. Kim, J.-S., Wu, H., Sweet, T. M., Molenaar, D., Junker, B. W., Moustaki, I., Harring, J. R., Bulut, O., Tong, X. (C.), **Wallin, G.**, & Di Plinio, S. (Eds.). (2025). Proceedings of the International Meeting of the Psychometric Society: The 89th Annual Meeting, Prague, Czech Republic, 2024..

Book Reviews

1. **Wallin, G**. (2025). It's Up for Debate: A Review of *Seminal Ideas and Controversies in Statistics* by Roderick J. A. Little. Accepted for publication in *Journal of Educational and Behavioral Statistics*.

2. **Wallin, G**. (2024). Review of *An Introduction to R and Python for Data Analysis: A Side-By-Side Approach* by Taylor R. Brown. *The American Statistician*.

Submitted Manuscripts

1. **Wallin, G**. & Huang, Q. (2026). A Hybrid Latent-Class Item Response Model for Detecting Measurement Non-Invariance in Ordinal Scales. arXiv:2601.17612. Under review at *British Journal of Mathematical and Statistical Psychology*.

2. Ma, Y*., Ushakova, A., Cain, K., & **Wallin, G**. A Statistical Framework for Dynamic Cognitive Diagnosis in Digital Learning Environments. Submitted to the *Journal of the Royal Statistical Society - Series A*. Major Revision. *Student author under my supervision.

Work in Progress

1. **Wallin, G.** & Huang, Q. DIF Analysis with Unknown Groups and Anchor Items for Asymmetric IRT Models. *British Journal of Mathematical and Statistical Psychology*, Special Issue on Asymmetric IRT. Invited to submit a full manuscript after editorial review of 1,000-word proposals; full paper due April 2026.

2. **Wallin, G.** & Bhaktha, N. Change-Point Analysis for Response-Time Modelling. Planned submission to *Psychometrika* in the first quarter of 2026.

3. Yawen, M., Ishida, S., Cain, K. & **Wallin, G.** NLP-Informed Diagnostic Statistical Models. *Psychometrika*. Invited to submit a full manuscript after editorial review of 1,000-word proposals; full paper due April 2026.

Grants and Scholarships

- Workshop funding from Institute of Mathematics for Industry (IMI), Kyushu University. 700,000 Yen. 2026
- Duolingo Competitive Research Grant. $60,000 USD. 2025
- Workshop funding from the STOR-i CDT. Lancaster University. £5,000 2025
- PhD studentship funding from the ESRC and NWSSDTP.
  (Awarded funds declined due to financial uncertainty in the NGO sector where external partner operates.) 2024
- Workshop funding from the Data Science Institute. Lancaster University. £5,000 2024
- LU ODA Early Career Fellowship. Lancaster University. £6,000 2024
- Research Award. Umeå School of Business, Economics and Statistics. 10,000 SEK 2022
- International Postdoc. Swedish Research Council. 3,150,000 SEK 2021
- Research Scholarship. Faculty of Social Science, Umeå University. 40,000 SEK 2019
- Research Scholarship. Kempe Foundation. 25,000 SEK 2018
- Research Scholarship. Umeå School of Business, Economics and Statistics. 40,000 SEK 2018
- Travel Grant. Wallenberg Foundation. 4,200 SEK 2018

- Travel Grant. Kempe Foundation. 5,000 SEK 2017
- Research Scholarship. The Student union at Umeå University. 8,500 SEK 2017
- Travel Grant. Wallenberg Foundation. 8,000 SEK 2017
- Travel Grant. Faculty of Social Sciences at Umeå University. 8,000 SEK 2016
- Research Scholarship. The Student union at Umeå University. 10,000 SEK 2016

Total amount awarded (in GBP, approx.): £326,000

Professional Activities

- **Member of Advisory Board** 2026-
  Swedish National Agency for Education.
- **Co-Editor** 2025-
  Proceedings of the International Meeting of the Psychometric Society.
- **Associate Editor** 2025-
  Teaching Mathematics and Its Applications.
- **Consulting Editor & Editorial Board Member** 2024-
  British Journal of Mathematical and Statistical Psychology.
- **Executive Committee member** 2024-
  STOR-i Centre for Doctoral Training, Lancaster University:
  Responsibilities: Year-1 cohort, annual spring school, annual conference, associates scheme, workshop funding, international visitor's programme, seminar series.
- **Secretary** 2024-
  Young Statisticians Section, Royal Statistical Society.
- **Website & Awards committee member** 2023-
  Psychometric Society.
- **Journal Reviewer**
  Annals of Applied Statistics
  Journal of the Royal Statistical Society - Series A
  Journal of Multivariate Analysis
  The American Statistician
  Statistics and Probability Letters
  Psychometrika
  Computational Statistics and Data Analysis
  British Journal of Mathematical and Statistical Psychology
  Psychological Methods
  Applied Psychological Measurement
  Behavior Research Methods
  TEST
  Educational and Psychological Measurement
  Discover Education
- **Program Committee Member** 2024
  International Meeting of the Psychometric Society (IMPS). Prague, Czech Republic.
- **Student Mentor** 2024-2025
  International Meeting of the Psychometric Society (IMPS). Prague, Czech Republic.
- **Reviewer** 2023
  National Council on Measurement in Education, Annual Meeting. Chicago, United States.
- **Local Organizing Committee Member** 2022
  Institute of Mathematical Statistics (IMS), Annual Meeting. London, UK.

- **Reviewer for PhD travel award** 2022
  International Meeting of the Psychometric Society
- **Member of the jury for best poster presentation** 2022
  International Meeting of the Psychometric Society
- **Conference Chair** 2020–2025
  International Meeting of the Psychometric Society
  National Council on Measurement in Education
- **Board Member** 2018–2020
  Umeå School of Business, Economics and Statistics
  The Graduate Student Association at the Faculty of Social Sciences at Umeå University
- **Committee member** 2018–2019
  The Graduate Student Committee of the Psychometric Society

Supervision and Mentorship

- **PhD Main Supervisor**, Lancaster University 2025-
  Student: Jasmine Burgess
- **PhD Co-Supervisor**, Lancaster University 2024-
  Students: Fay Benedik, Yawen Ma
- **MSc Supervisor**, Lancaster University 2024-2025
  Student: Jasmine Burgess
  Jayven Bahia
  Junhao Guo
  Elizabeth Mwania (African Institute of Mathematical Sciences)
  Joe Rutherford
  Kristina Grolmusova
- **Mentor for early career reviewers** 2025–
  British Journal of Mathematical and Statistical Psychology
- **Supervisor for Summer Intern**, Lancaster University 2024
  Student: Anais La Chon
- **Academic Advisor, Lancaster University** 2023 –
  Advisor to around 15 MSc Data Science students per year.
- **Bachelor Thesis Supervisor, Umeå University** 2018
  Thesis topic: Kernel smoothing techniques with applications to educational testing.
  Students: Jonas Jonsson; Samuel Brändström
- **Master Thesis Supervisor, Umeå University** 2015 –2020
  Yearly quantitative advisor to business students writing their master's thesis.

Awards and nominations

- **Invitational Fellowship - Japan Society for the Promotion of Science** 2025
  Includes a 6-week research visit to the University of Tokyo.
- **Best Reviewer Award** 2025
  Psychometrika
- **Recipient of the Umeå School of Business, Economics and Statistics Research Award** 2022
  Umeå University
- **Recipient of the LSE Class Teacher Award** 2022
  London School of Economics and Political Science

- **Pedagogical Prize Nomination** 2017
  Umeå School of Business, Economics and Statistics

Organisation

- **Symposium: Advances in Modeling and Detecting Complex Patterns of Measurement Non-Invariance.**
  Invited international speakers. 2026
  International Meeting of the Psychometric Society
- **Workshop: Contemporary Challenges in Change-point Analysis.**
  Invited speakers from the UK and overseas. 2026
  Lancaster University
- **Workshop: Innovations in Social Data Science - Tackling Inequality in Resource-Constrained Contexts with Novel Data and Methods.**
  Invited speakers from Oxford, Southampton, GRADE (Peru), Save the Children, and Lancaster University. 2025
  Lancaster University
- **STOR-i Spring School 2025-26.**
  A week-long event for Lancaster PhD students, in collaboration with NGO partner Nesta (2025) and Save the Children (2026). 2025-26
  Lancaster University
- **STOR-i Annual Conference 2025-26.**
  A three-day conference for our CDT students, with a programme including external, international academics, alumni and collaborating partners. 2025
  Lancaster University
- **Data on the Shire 2024.**
  A week-long event for Lancaster PhD students together with an Irish CDT (¿100 participants), and in collaboration with the British Red Cross. 2024
  Lancaster University

Teaching and Examination

- **Internal Examiner, PhD thesis** 2025-2026
  Owen Li, Ophelie Schaller, Lancaster University
- **Lecturer and Course Responsible** at Lancaster University 2023-
  MSc: Statistical Learning, Unsupervised Learning, Ethics and Communication. BSc: Project Skills.
- **Lecturer and Course Responsible** at African Institute for Mathematical Sciences, Cameroon 2024-2025
  Data mining and Unsupervised Learning.
- **Guest Lecturer** at the London School of Economics 2024
  MSc-level course on quantitative methods at the Dept. of Sociology.
- **Lecturer and Teaching Assistant** at The London School of Economics 2021-2024
  Multivariate Analysis and Measurement (Lecturer)
  Statistical Models and Data Analysis (TA)
  Sample Surveys and Experiments (TA)
  Elementary Statistical Theory I (TA)
- **Lecturer and Course Responsible** at Umeå University 2018
  Sampling: Design and Analysis
- **Lecturer** at Umeå University 2017–2019
  Probability and Statistical Inference
- **Lecturer** at Umeå University 2017–2019
  Basic Statistics

- **Teaching Assistant** at Umeå University 2017
  Statistics for Journalists
- **Teaching Assistant** at Umeå University 2014–2017
  Basic Statistics

Conference Talks and Invited Presentations

Statistics Seminar Series Queen Mary University of London, UK
Invited talk 2026

Seminar on Psychometrics and Latent Variable Models Pontificia Universidad Catolica, Chile
Invited talk 2026

Statistics Seminar Series Linköping University, Sweden
Invited talk 2025

15th Scientific Meeting of the Classification and Data Analysis Group Naples, Italy
Contributed talk. 2025

International Meeting of the Psychometric Society Minneapolis, US
Contributed talk. 2025

Institute of Mathematics and Related Sciences Lima, Peru
Invited talk. 2025

Group for the Analysis of Development Lima, Peru
Workshop. 2025

Group for the Analysis of Development Lima, Peru
Invited talk. 2025

Statistics Seminar Series University of Edinburgh, UK
Invited talk. 2025

UK Rasch and Psychometrics User Group Meeting 2025 Cambridge, UK
Contributed talk. 2025

International Meeting of the Psychometric Society Prague, Czech Republic
Contributed talk. 2024

Data Science Seminar Series Lancaster University, UK
Invited talk. 2024

Statistics Seminar Series Lancaster University, UK
Invited talk. 2023

International Meeting of the Psychometric Society Maryland, USA
Contributed talk. 2023

European Congress of Methodology Ghent, Belgium
Contributed talk. 2023

Nordstat conference Gothenburg, Sweden
Contributed talk. 2023

Annual Meeting of the National Council on Measurement in Education Chicago, USA
Contributed talk. 2023

Social Statistics Research Seminar London, LSE
Invited talk. 2022

Computational Psychometrics Research Seminar Prague, Czech Republic
Invited talk. 2022

International Meeting of the Psychometric Society Bologna, Italy
Contributed talk. 2022

Cramér Society Research Seminar Sweden (Virtual)
Invited talk. 2022

Social Statistics Research Seminar London, LSE
Invited talk. 2022

International Meeting of the Psychometric Society Virtual
Contributed talk. 2021

Computational Psychometrics Research Seminar Prague, Czech Republic (Virtual)
Invited talk. 2021

International Meeting of the Psychometric Society Virtual
Contributed talk. 2020

MAASAI Research Seminar Nice, France
Invited talk. 2020

International Meeting of the Psychometric Society Santiago, Chile
Invited talk. 2019

Frontier Research in Educational Measurement Oslo, Norway
Contributed talk. 2018

International Meeting of the Psychometric Society Zürich, Switzerland
Contributed talk. 2017

Annual Meeting of the National Council on Measurement in Education San Antonio, the US
Contributed talk. 2017

The Winter Conference in Statistics Åre, Sweden
Contributed talk. 2017

International Meeting of the Psychometric Society Asheville, the US
Contributed talk. 2016

Pedagogical Courses and Other Training

- **Postgraduate Certificate in Educational Practice (PGCEP)** at Lancaster University
  2-year programme. Finish in June 2026.

  ED&I Training, at Lancaster University. 2024, 2025.
- **Teaching in higher education** at Umeå University

3 ECTS (2 weeks full-time)

- **Learning in higher education** at Umeå University
  4.5 ECTS (3 weeks full-time)
- **Didactics for university teachers** at Umeå University
  3 ECTS (2 weeks full-time)